TABLEAU DATA ANALYST - ADVANCED PRACTICE PAPER 3

ANSWER KEY WITH EXPLANATIONS

QUICK ANSWER KEY

Q1: B, C	Q11: D	Q21: A	Q31: D	Q41: B, D, E	Q51: A
Q2: A	Q12: D	Q22: B, D	Q32: A-B-C-D	Q42: D	Q52: C
Q3: D	Q13: C	Q23: A-B-C-D-E	Q33: C, D	Q43: A-B-C-D	Q53: B
Q4: A-B-C-D-E	Q14: A-B-C-D	Q24: C	Q34: A	Q44: B	Q54: B
Q5: D	Q15: A	Q25: B	Q35: C	Q45: A	Q55: C
Q6: B	Q16: B	Q26: A, C	Q36: A, B, E	Q46: C	Q56: D
Q7: A, C	Q17: A	Q27: D	Q37: D	Q47: A	Q57: A
Q8: A	Q18: B	Q28: A	Q38: B, C, E	Q48: C	Q58: B
Q9: B	Q19: C	Q29: B	Q39: C	Q49: D	Q59: A
Q10: C	Q20: D	Q30: A-B-C-D	Q40: A	Q50: B, D, E	Q60: D

Scoring: give yourself 1 point per question only when the answer is COMPLETELY right (all letters on multi-selects; the exact sequence on build lists). 39+/60 keeps you on track for the real exam's 65% pass mark; 45+ means exam-ready.

DETAILED EXPLANATIONS

Question 1 - Correct answer: B, C

Context filters move early in the pipeline - only extract and data source filters run before them - so on relational sources Tableau can materialize a temporary table of the context-filtered rows and reuse it. Because they run BEFORE FIXED LODs, promoting a filter to context changes what the LOD computes. Context filters never run after table calculations, and the temporary-table build can sometimes make queries SLOWER, not faster - an improvement is never guaranteed.

Question 2 - Correct answer: A

Tableau Desktop has a dedicated PDF File connector that scans the document and imports the tables it detects, each as a separate table. Data Interpreter cleans messy Excel/Google Sheets, not PDFs; the Text File connector cannot parse PDF structure; and there is no image-to-data shortcut for local files.

Question 3 - Correct answer: D

The FIXED LOD stamps 300 (East's true total) onto BOTH of East's underlying category rows. Region alone is in the view, so those two rows aggregate into the single East mark and SUM adds the stamped values: 300 + 300 = 600. To avoid the inflation, aggregate the LOD field with AVG, MIN, MAX, or ATTR - each returns 300 for East.

Question 4 - Correct answer: A-B-C-D-E

The set must exist before anything can reference it (A). The comparison sheet is wired to the set's In/Out membership and the sheets assembled on a dashboard (B). Only then can the Change Set Values action be added and aimed at the existing set (C), configured so clicks ADD members and clearing the selection keeps them (D), and finally tested (E).

Question 5 - Correct answer: D

Sales total 230 and there are 2 DISTINCT orders, so SUM([Sales]) / COUNTD([Order ID]) = 230 / 2 = 115. AVG([Sales]) and SUM/COUNT both divide by the 3 underlying LINES (230 / 3 = 76.7) because COUNT counts every non-null row, not distinct keys - the classic COUNT vs COUNTD trap.

Question 6 - Correct answer: B

Tableau Catalog (a Data Management component) builds lineage across the site, letting owners trace which dashboards, workbooks, flows, and data sources consume a table or field before making breaking changes. The Optimizer tunes workbook performance, Bridge tunnels refreshes to on-prem sources, and data quality warnings only flag known issues - none of them map dependencies.

Question 7 - Correct answer: A, C

FIRST() is the offset from the current row to the FIRST row, so it is 0 on the first row itself. SIZE() counts the rows in the partition. The traps: LAST() is the offset to the LAST row - it is 4 on the first row of five but 0 on the last row - and INDEX() is the current position and starts at 1, never 0.

Question 8 - Correct answer: A

Nested LODs evaluate inside-out. The inner FIXED per Region produces 300, 120, 50; the outer FIXED has NO dimensions, so it computes one value over the whole data source - the MAX of those region values, 300 - and stamps it on every row regardless of what is in the view. 470 would only be the SUM of regions, which MAX does not do.

Question 9 - Correct answer: B

Performance Recording (Help > Settings and Performance) records each step's duration while you interact with the workbook and opens a summary workbook ranking query, render, and layout times - built for exactly this diagnosis. The Optimizer audits published workbooks, and Catalog and Data Interpreter perform governance/cleaning roles, not timing.

Question 10 - Correct answer: C

Prep's profile pane lets you select the null bars and Fill Null with a fixed value (0 here) inside a Clean step - targeted and reversible. Aggregate reshapes to grouped totals, Pivot rearranges columns, and Union stacks rows; none of them substitute values for nulls.

Question 11 - Correct answer: D

'asc' ranks SMALLEST values first: 200 gets rank 1, the tied pair of 300s share competition rank 2, and 500 gets rank 4 (gaps are kept around ties). Every other practice question in this series used the default 'desc' - examiners flip the argument precisely to catch memorized patterns.

Question 12 - Correct answer: D

The Trend Lines options dialog includes 'Allow a trend line per color', which fits one model per color group so every category gets its own fitted line and statistics. Trend lines CAN respect color; forecasting does not create per-group fit lines; and moving the field to Detail alone does not produce per-line models.

Question 13 - Correct answer: C

WINDOW_MAX scans the entire partition (no offset arguments given, so the whole table) and returns its largest value - 30 - on every mark, including March. This constant stamping is what lets you write WINDOW_MAX(SUM([Sales])) - SUM([Sales]) to measure each month's distance from the best month (March would show 10, February 0).

Question 14 - Correct answer: A-B-C-D

Alerts attach to a continuous numeric axis of a published view, so first open that view (A), create the alert from the axis (B), configure operator/threshold/check frequency (C), and finally add recipients and save (D). Without a numeric axis - e.g., on a filled map - the Create Alert option is unavailable.

Question 15 - Correct answer: A

Initial SQL is sent by the database server the moment a live connection opens. With an extract, Tableau queries the local .hyper file and no database session exists, so Initial SQL is unavailable. Extracts do preserve their filters and metadata; privileges are irrelevant here.

Question 16 - Correct answer: B

Measure Values stacks all qualifying measures into one pill that plots each as its own mark line, with Measure Names distinguishing (color or axis header) which is which - the standard multi-measure single-chart build. Triple dual axes are unnecessary and unmanageable, stacking merges bars, and blending has nothing to do with multiple measures of the same source.

Question 17 - Correct answer: A

DATEADD adds calendar units and clamps overflow days to the target month's last day. 2025 is not a leap year, so February 29 cannot exist and the result becomes February 28, 2025. (The same clamping rule makes January 31 + 1 month = February 29 in a leap year - Tableau's documented behavior, never an error.)

Question 18 - Correct answer: B

Right-clicking a step and choosing 'Add Branch' creates a second lineage from that point, after which each side can apply its own filters and outputs - one flow, two outputs, one shared cleaning chain. Union adds rows to one table, maintaining two flow copies duplicates logic and invites drift, and Aggregate performs grouping, not branching.

Question 19 - Correct answer: C

Published workbooks keep a version history on Server/Cloud; administrators and content owners can view it and restore any earlier version, reverting the content in place. The Recycle Bin is for deleted (not historical) items, and the Optimizer is a performance auditing tool.

Question 20 - Correct answer: D

A FIXED LOD computes each customer's line count at customer grain, so the boolean filters customers (not rows or marks), and as a row-level field it works identically on every sheet. A Top N filter changes meaning as data changes and ranks, not thresholds; ranking table calcs behave at the view level; and an extract filter would drop rows before any counting.

Question 21 - Correct answer: A

The Workbook Optimizer scans published workbooks (web authoring) for anti-patterns - hidden sheets, unused fields, too many quick filters, un-optimized sources - and applies recommended performance fixes with your approval. The Performance Recorder only measures timing on a workbook you open; Bridge tunnels data; Data Interpreter cleans spreadsheet noise.

Question 22 - Correct answer: B, D

A data source filter is defined once on the data source and travels with it - every user, sheet, and workbook using that source inherits it. In the pipeline it sits between extract filters (earlier) and context filters (later). The traps: extract AND context filters can also change FIXED LOD results - a data source filter is plainly not the only one - and no data source filter can ever run after table calculations: data source filters are among the earliest operations, table calc filters the very last.

Question 23 - Correct answer: A-B-C-D-E

A Pareto needs descending bars first (A, B), because the cumulative percent is only meaningful over the descending order. The running total accumulates sales (C), the secondary Percent of Total converts that accumulation into the familiar cumulative share (D), and the dual-axis line overlays the curve on the bars (D), finishing with validation of the ~100 percent endpoint (E).

Question 24 - Correct answer: C

Row-level ratios are 100/1 = 100 and 200/4 = 50, and their average is (100 + 50) / 2 = 75 - the 'average of ratios'. The aggregated, business-weighted ratio would be SUM/SUM = 300 / 5 = 60 instead. Multiplying the averages gives 375 and SUM of products gives 900 - both meaningless here. Exams love this row-level versus aggregate distinction.

Question 25 - Correct answer: B

Sparklines are just very small multiple line charts (product on Rows, continuous date on Columns, measure on the axis) with everything decorative removed - each axis's right-click menu offers Show Header, which you clear. Fit options only scale the view, deleting the measure would delete the line, and covering axes with objects is not hiding them.

Question 26 - Correct answer: A, C

Since Tableau 2020.1, extracts store the logical and physical tables of a multi-table model at their own granularities, and incremental refresh keys on a date or incrementing column to append only new rows. Credentials live with the published connection - not inside the extract file - and extract filters are fully editable in the Extract Data dialog.

Question 27 - Correct answer: D

Lifetime value exists at customer grain, so per-customer totals must come from a FIXED LOD (Anna 120, Ben 300, Cara 80, Dan 100), then a CASE turns them into bucket labels used as a dimension; COUNTD counts distinct customers per bucket (High = 1, Low = 3). A row-level sales filter would drop Ben's 100 line (not the customer), Top 10 has nothing to do with value thresholds, and EXCLUDE is for dimensions in the view.

Question 28 - Correct answer: A

Relative to Next compares each mark against the FOLLOWING mark: April = (140 - 175) / ABS(175) = -35 / 175 = -20%. The 25% trap flips the division to (175 - 140) / 140 (that would be May's relative-to-previous value); +20% ignores the direction; and NULL belongs to May, which has no next mark to compare against.

Question 29 - Correct answer: B

Virtual Connections centralize the database connection (credentials and settings live on the server, not in workbooks), and their data policies enforce row-level rules for every connected workbook - exactly connection governance plus invisible row-level security. Certification only endorses quality, Bridge tunnels refreshes, and per-workbook filters require editing (and trusting) every workbook.

Question 30 - Correct answer: A-B-C-D

The controlling construct starts with the user-facing control - the parameter (A). The boolean calculation translates the parameter into a true/false test (B). The zone is then bound to that boolean via Layout > 'Control visibility using value' (C), and finally the control is exposed and validated (D). A boolean controlling field that returns a single value for the view is the requirement for dynamic zone visibility.

Question 31 - Correct answer: D

REGEXP_EXTRACT returns the FIRST substring matching the pattern. The first run of consecutive digits in 'US-2024-1186' is '2024'; the trailing '1186' would need a different pattern (for example a capture group anchored to the end). 'US' contains no digit, and a non-match would return NULL - but here the pattern does match.

Question 32 - Correct answer: A-B-C-D

Percent of Total first converts each month to its share of the partition's full sales (B); the secondary Running Total then ACCUMULATES those shares (C), which is Tableau's standard way to stack two table calculations. The view itself is just the monthly line (A), and the endpoint check (D) confirms the composition - 100 percent at the end of the partition.

Question 33 - Correct answer: C, D

Groups are manual, within-dimension labelers: drag members together (or use the paperclip in the view) and the rest can drop into an 'Other' bucket. They never re-evaluate themselves (that is what dynamic SETS do) and a single group belongs to exactly one dimension, so cross-dimension members cannot be folded into one field.

Question 34 - Correct answer: A

When the IIF test evaluates to unknown (a NULL date compared with TODAY()), neither the then-branch nor the else-branch fires: IIF returns its optional FOURTH 'unknown' argument, or NULL when it is omitted - so these rows show blank. Adding the fourth argument supplies an explicit 'Unknown'. Nothing errors, and a NULL date never compares as smaller or larger, so both branch readings are myths.

Question 35 - Correct answer: C

Each FIXED partition (State B, State C) contains only that state's own rows, so no filtered-away row could ever contribute to a surviving partition - the values are 200 and 300 with a plain filter, and still 200 and 300 after the filter becomes context. Context re-computation only changes LOD results when removed rows belong to a partition that STAYS (for example, filtering Category of a Region-level LOD).

Question 36 - Correct answer: A, B, E

Reference lines/bands offer Table, Pane, and Cell scoping; distribution bands support percentile (and standard deviation) definitions; and a forecast cannot exist without a date and a measure. The two decoys reverse real facts: box plots and trend lines live in the ANALYTICS PANE, not on the Marks card (Shape builds shape encodings, not statistics).

Question 37 - Correct answer: D

REPLACE substitutes EVERY occurrence of the search string: each of the three 'a' characters in Banana becomes 'o', giving B-o-n-o-n-o = 'Bonono'. Replacing only the first occurrence is a common misconception - Tableau has no 'first-only' flag, so 'Bonana' and 'Banano' are traps.

Question 38 - Correct answer: B, C, E

Extract refreshes, published Prep flow runs (via Tableau Prep Conductor), and subscriptions are all scheduled, unattended operations. Alerts are condition-driven: they fire the MOMENT a threshold is crossed and never on a calendar, and custom views are personal saved states with no delivery mechanism - so those two claims are false.

Question 39 - Correct answer: C

The FIXED LOD gives each region mark its total (East 300, South 120, West 50), and the descending rank table calculation numbers them by that value: East = 1, South = 2, West = 3. Ranking an LOD measure is a common exam pattern for 'rank by total, even when finer details are filtered' - the numbers, not the visible marks, drive the rank.

Question 40 - Correct answer: A

DATETRUNC snaps a date to the FIRST instant of the specified part. May 20 falls in the second quarter (April-June), so the quarter floor is April 1, 2024. May 1 is the MONTH floor, July 1 the NEXT quarter's start, and January 1 the YEAR floor - each a different truncation level.

Question 41 - Correct answer: B, D, E

Blue/discrete fields produce category headers; green/continuous fields produce axes - and dates and numbers can be either, switched by drag or right-click convert. A continuous measure on Color creates a gradient legend. Conversions change only how MARKS are drawn - never the stored data - and dimensions absolutely can be continuous, for example a continuous numeric dimension drawn as an axis.

Question 42 - Correct answer: D

BUFFER(MAKEPOINT(lat, lon), distance, units) builds a radius polygon around the point - it is exactly the field needed for delivery-zone rings. DISTANCE returns a scalar length between two points, MAKELINE joins two points with a line (it takes two points, not a radius), and the last option reverses the argument order and swaps lat/long, which would misplace every zone.

Question 43 - Correct answer: A-B-C-D

Row-level security starts with the user-function calculation (A - USERNAME picks up the signed-in user, ISMEMBEROF their group membership). The boolean goes on Filters kept True (B), is broadcast to all sheets using the data source so no sheet escapes it (C), and is finally verified per user or with the filter card's temporary 'View as' feature (D).

Question 44 - Correct answer: B

INCLUDE adds Order ID beneath Region: first each ORDER's sales is totaled (O1 = 100 + 50 = 150; O2 = 80), then the view's aggregation rolls those order totals up: SUM(150, 80) = 230. AVG would show 115, MIN 80, and 150 is only O1 - the aggregation you wrap the LOD in changes the answer, which is why the exam names it explicitly.

Question 45 - Correct answer: A

FIXED LODs are computed before measure filters, just as they are before dimension filters. The measure filter removes West's 500-row from the view entirely (its mark vanishes), but East's rows - 60 and 40, both under the limit - remain, and the LOD still returns East's full region total of 100. Measure filters change WHICH MARKS exist, never an already-computed FIXED value.

Question 46 - Correct answer: C

Cell addressing makes EVERY mark its own partition, so each cell's percent-of-total denominator is the cell itself: 100/100, 200/200, 300/300 - all 100%. The useful reading (Table down) would be 16.7/33.3/50 per column; 30/60/100 invents a cumulative share. Choosing between Table, Pane, and Cell changes the denominator - exactly what exams test.

Question 47 - Correct answer: A

Table calculation filters are applied LAST and only hide marks - they never recompute anything underneath. The FIXED grand total was already 600 across the full data (100 + 200 + 300), stamped on every mark, so the surviving March mark still shows 600. A dimension or measure filter also leaves FIXED untouched; only a CONTEXT filter (or extract/data source filter) would change it.

Question 48 - Correct answer: C

Fit Wide stretches the view horizontally to fill the available width (height unchanged); Fit Tall is its vertical twin; Fit Entire View rescales in BOTH directions; Standard reverts to the default canvas. Picking the wrong Fit is a classic dashboard-layout mistake.

Question 49 - Correct answer: D

Relationships keep each table at its own grain and combine aggregates at query time - ideal for SUM/SUM style measures, but row-level math mixing fields of both tables across a relationship can produce surprising aggregations. When a calculation needs both tables' fields row by row, the documented recommendation is a physical join, which materializes one merged table and makes row-level math exact. Blends aggregate after the fact, and LODs cannot fix cross-table row-level logic.

Question 50 - Correct answer: B, D, E

The Data Management add-on unlocks Prep Conductor, Catalog, and Virtual Connections (plus extended Bridge support). Subscriptions and custom views are core platform features that licensed users get with no add-on - exactly where the decoy options sit.

Question 51 - Correct answer: A

FIXED dimensions treat NULL as just another member: the LOD builds a partition for the NULL level and computes its own sum, exactly as it would for 'East' or 'West'. If you want nulls folded elsewhere, handle them first (for example with IFNULL) - but by default they are neither dropped nor inflated to the grand total, and nothing errors.

Question 52 - Correct answer: C

Prep's Clean step re-types fields on demand; when text like 'N/A' cannot parse as a number, Prep flags it invalid and converts it to NULL rather than erroring, after which Fill Null (or Remove) handles it explicitly. Renaming changes a label, aggregation is view-level behavior, and pivoting reshapes columns without ever converting types.

Question 53 - Correct answer: B

MIN and MAX are valid aggregates on strings, where they compare ALPHABETICALLY: the character '9' sorts after '1', so '99' is the maximum and '100' the minimum - the classic string-number inversion. Converting with INT() before comparing would yield the numeric answer, but the raw MAX on text answers '99'.

Question 54 - Correct answer: B

MIN and MAX are aggregates that work on dates: MIN returns the EARLIEST date (2024-01-05), MAX the latest. This is the foundation of first-purchase LODs - {FIXED [Customer] : MIN([Order Date])} - and it needs no extra date functions as long as the field is a true date rather than text.

Question 55 - Correct answer: C

Every parameter action exposes a 'Clearing the selection will' choice: Keep the current value, or Set value to a specified target (a constant, the parameter's default, or a null-equivalent for the parameter's type) - the latter resets the dashboard on deselect. Context filters and filter actions operate on entirely different mechanics.

Question 56 - Correct answer: D

A set can be surfaced like a filter card using 'Show Set': its panel lists current members and authorized users add or remove names interactively (fixed sets only - dynamic sets recalculate themselves instead). Set actions change membership through clicks on MARKS, not through a member panel, and groups cannot be edited at runtime.

Question 57 - Correct answer: A

The union dialog's wildcard matches TABLES AND WORKSHEETS of the current connection - matching sheet names in the same workbook stacks their rows and generates a Table Name column identifying each sheet's origin. Blends and joins operate across differently structured rows/columns, and the wildcard reaches inside one file just as easily as within a folder.

Question 58 - Correct answer: B

Import Custom Geocoding assigns your own geographic values (region names, depot codes) to latitude/longitude, teaching Tableau about places its built-in database lacks. Edit Locations only remaps values against BUILT-IN locations, MAKEPOINT plots coordinate pairs rather than fixing name geocoding, and Data Interpreter cleans spreadsheet headers.

Question 59 - Correct answer: A

DATENAME returns the date part's NAME as a string - here 'Wednesday'. The integer 3 is what DATEPART('weekday', ...) returns, and abbreviated names are not DATENAME's output. Remember the pairing: DATEPART = number, DATENAME = name.

Question 60 - Correct answer: D

ZN is 'zero-null': strictly numeric, substitution value fixed at 0. IFNULL is the general tool - any field type, any replacement value (a string, a date such as TODAY(), another field). That asymmetry is a favorite trap: ZN([Country]) fails because Country is text, while IFNULL([Country], 'Unknown') succeeds.